

SafeSite: AI-Driven Safety Compliance

Automated PPE Detection for Safer Workplaces

Group Members: Ian, Peter, Troy

Project Tier: 1



THE PROBLEM

What is the problem?

Construction sites are too large for manual oversight. It's impossible to watch every worker at all times to ensure they are wearing safety gear.

Who cares?

Construction companies, site supervisors, OSHA, and site safety officers.

Why is it important?

Missing PPE leads to severe, preventable head injuries and accidents.

Financially, non-compliance costs companies over \$15,000 to \$50,000 per violation in OSHA fines and dramatically raises insurance premiums.

SOLUTION OVERVIEW

Automated PPE checking system using object detection.

Instead of manual checks, our system automatically scans site camera photos to find workers missing mandatory safety gear.

How it works:

- Camera takes site photos
- YOLO model detects workers and gear
- System flags missing helmets/vests
- Generates compliance report



TECHNICAL APPROACH



Computer Vision Technique

Object Detection. We need to locate multiple specific items (helmets, vests, people) within a single image, making object detection the right choice over simple classification.



Model

YOLOv8. It is highly documented, easy to fine-tune, and is the industry standard for fast, real-time object detection.



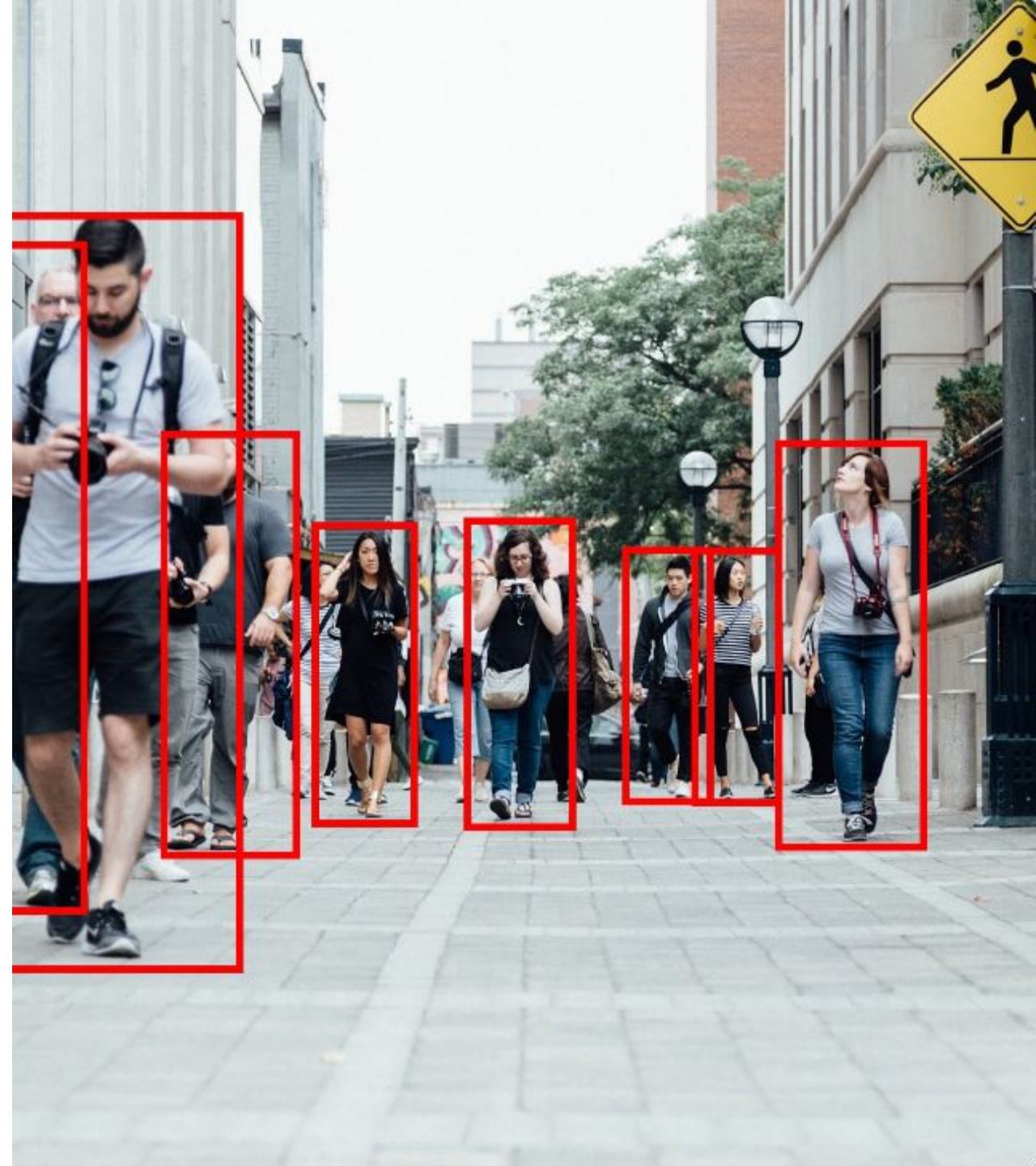
Framework

PyTorch & Ultralytics. Ultralytics provides a streamlined, built-in pipeline specifically designed to train and run YOLO models efficiently.

DATA PLAN

Source: Roboflow Universe (Public Construction PPE Dataset).

- **Size:** Approx. 1,200 to 1,500 samples.
- **Labels:** Helmet, No-Helmet, Vest, No-Vest
- **Preparation:** The dataset is already labeled. We will apply auto-resizing to 640x640 to match YOLO's expected input and clean out broken images.



SYSTEM DIAGRAM



[Input]
Site Camera

Photo
(Image
Acquisition)



[Processing 1 & 2]
YOLOv8 Inference

(Detect Person → Detect
PPE)



[Output]
Flagged

Image
(Visual bounding boxes)

SUCCESS METRICS

$\geq 85\%$

PRIMARY: MAP ACCURACY

$\sim 60\text{ms}$

SECONDARY: SPEED (GPU)

Our primary goal is high accuracy to avoid false alarms. For speed, we target $\sim 60\text{ms}$ per image on a GPU (which translates to $\sim 800\text{ms}$ on a standard CPU), ensuring it's fast enough for periodic site snapshots.



WEEK-BY-WEEK PLAN

Timeline	Task	Milestone
Week 1	Get Roboflow dataset & set up Colab environment	Dataset Ready
Week 1	Train initial YOLOv8 model & evaluate metrics	Model Working
Week 2	Test model, add augmentations, improve accuracy	Good Accuracy
Week 2	Run inference on test images to create final visual demo	Demo Ready
Week 2	Final presentation, testing, and GitHub documentation	Everything Done

CHALLENGES & BACKUP PLANS

Challenge: Low Accuracy

The model might struggle to differentiate safety vests from standard reflective clothing or regular shirts.

Plan B: Add image augmentations (like random crop or brightness adjustments) during training to help the model generalize better.

Challenge: Not Enough Data

A single dataset might lack variety in lighting or construction site backgrounds.

Plan B: Switch to or merge multiple different PPE datasets directly from Roboflow Universe to boost the training volume.

RESOURCES NEEDED



Compute

Google

Colab

Using the free GPU tier for training the model quickly.



Frameworks

PyTorch & Ultralytics

Standard Python libraries for deep learning and YOLO models.



Estimated Cost

\$0

Total

Using free public datasets and free open-source software.